Pooling idempotents and the Beck defects of the absolute tower

A corrected form of the shadow decomposition

Vinay K. Awasthi
Department of Applied Mathematics and Statistics, Johns Hopkins University

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Abstract

The two candidate mixed distributive laws of the absolute three-level tower [1] fail some Beck axioms. We show that every defect of the shift law is governed by one natural transformation, the pooling idempotent $e = \eta_{\mathsf{D}} \circ \mu \colon \mathsf{D}^2 \Rightarrow \mathsf{D}^2$, which sends a law on laws to the point mass at its barycenter. At level k the unit axiom holds up to the nested pooling $P_k = \eta \circ \mathsf{D}^{k-1}(\mu)$: the left side is P_k applied to the right side, and the right side is reached through k stages, one pooling per depth. The multiplication axiom fails by the same P_k, applied inside a mixture instead of outside it. Because e is not affine, the two defects point in opposite directions, which is the structural reason behind the tower paper's result that no partial order orients both. Every statement is a one-line consequence of the monad laws and is verified in exact arithmetic, including on a three-point base. This note replaces an earlier draft, the "shadow decomposition", whose 3-cells were nearest-atom quantisers; we record what that draft got right, what it got wrong, and why the corrected objects are 2-cells of **Cat** with no 3-cells needed.

1 What this note replaces

An earlier draft proposed a "shadow decomposition" of the shift law's Beck defects into crossover maps χ_k, coherence 2-cells α_k and 3-cells β_k, said to be conditional expectations onto the σ-algebras generated by a context, and asserted the coherence axioms of a tricategory. Its scripts were audited for this note (section 8). Three findings decided the rewrite.

1. The maps β_k send each support point to its nearest atom in total variation, with ties broken by the order of a list. That is a quantiser, not a conditional expectation: it does not preserve barycenters, it depends on the listing of the atoms, and on a three-point base it fails to commute with pushforward along a Markov kernel in about half of all random cases.
2. It does not decompose the defect: in the draft's own two-atom example, β_1 does not send the right side of (U) to its left side.
3. The "coherence" checks in the scripts were recorded as true without being computed, and the first script's β_1 returns the empty distribution on the standard input, so its checks passed vacuously.

The draft's instinct was right: in the single-atom case β_1 does undo the unit defect, by pooling. The corrected object is the barycentric pooling map, and it decomposes every defect exactly.

2 Setup

We use the notation of [1]. D is the finite distribution monad with unit η (point mass) and multiplication μ (barycenter). The absolute tower is the context comonad $\mathsf{C}_3 X = X \times \mathsf{E}_3 X$ with

$\mathsf{E}_3X = \mathsf{D}X \times \mathsf{D}^2X \times \mathsf{D}^3X \times \mathsf{Sh}X$ and points (x, c), $c = (\rho, \Pi, \Omega, \Sigma)$ [1, Thm. 6.1, Cor. 6.4]. The shift law is

$$\lambda^{\sigma}_X(M) = \Big(\sum_i a_i\delta_{x_i},\ \sum_i a_i\delta_{\rho_i},\ \sum_i a_i\delta_{\Pi_i},\ \sum_i a_i\delta_{\Omega_i},\ \sum_i a_i(K^{[1]})_*\Sigma_i\Big)$$

for $M = \sum_i a_i\delta_{(x_i,c_i)}$, with $K^{[1]}(\theta,\rho) = (\theta, \delta_\rho)$; it is natural, satisfies (Co), and fails (U), (M) and (Cm) [1, Thm. 8.3]. The averaging law $\bar{\lambda}$ satisfies (U), (Co), (M) and fails only (Cm) [1, Thms. 8.6, 8.7]. The Beck axioms are read left to right as in [1, Def. 2.1]. Note that λ^σ is affine: each component is a mixture over the samples, so $\lambda^\sigma \circ \mu_{\mathsf{C}_3} = \beta \circ \mathsf{D}(\lambda^\sigma)$, where β is mixing in $\mathsf{C}_3\mathsf{D}X$ (componentwise barycenter, [1, §8]).

3 The pooling idempotent

Definition 3.1 (Pooling)**.** $e_Y = \eta_{\mathsf{D}Y} \circ \mu_Y \colon \mathsf{D}^2Y \to \mathsf{D}^2Y$ sends $\nu = \sum_j \nu_j\delta_{p_j}$ to $\delta_{\sum_j \nu_j p_j}$. For $k \geq 1$ the *nested pooling* on $\mathsf{D}^{k+1}Y$ is $P_k = e \circ \mathsf{D}(e) \circ \cdots \circ \mathsf{D}^{k-1}(e)$, pooling the innermost depth first. On the context component of $\mathsf{C}_3\mathsf{D}X$ let P act by P_k on the level-k component and by $(\theta, \nu) \mapsto (\theta, e(\nu))$ on each hypothesis of the shadow.

Lemma 3.2. *(i) e is a natural transformation $\mathsf{D}^2 \Rightarrow \mathsf{D}^2$, split by μ and η_{D}: $e = \eta_{\mathsf{D}} \circ \mu$ and $\mu \circ \eta_{\mathsf{D}} = \mathrm{id}$. In particular e is idempotent and preserves the barycenter, $\mu \circ e = \mu$.*
(ii) e commutes with $\mathsf{D}(k^{\#})$ for every Markov kernel k, since $k^{\#}$ is affine.
(iii) e is not affine: $e(\frac{1}{2}\delta_{\delta_0} + \frac{1}{2}\delta_{\delta_1}) = \delta_{(\frac{1}{2},\frac{1}{2})} \neq \frac{1}{2}\delta_{\delta_0} + \frac{1}{2}\delta_{\delta_1}$.
(iv) $P_k = \eta \circ \mathsf{D}^{k-1}(\mu)$. For $k \geq 2$, P_k is not idempotent in general.
(v) $e \circ \mathsf{D}(\eta) = \eta_{\mathsf{D}}$, and more generally $P_k \circ \mathsf{D}^k(\eta) = \eta_{\mathsf{D}^k}$.

Proof. (i) Naturality of η and μ; $\mu \circ \eta_{\mathsf{D}} = \mathrm{id}$ is a monad law, so $e \circ e = \eta_{\mathsf{D}}(\mu\eta_{\mathsf{D}})\mu = e$ and $\mu e = \mu$. (ii) $\mu \circ \mathsf{D}(k^{\#}) = k^{\#} \circ \mu$ because $k^{\#}$ is affine, and $\mathsf{D}(k^{\#})$ maps point masses to point masses. (iii) Direct. (iv) By induction, $P_k = e \circ \mathsf{D}(P_{k-1}) = \eta\mu \circ \mathsf{D}(\eta) \circ \mathsf{D}^{k-1}(\mu) = \eta \circ \mathsf{D}^{k-1}(\mu)$, using $\mu \circ \mathsf{D}(\eta) = \mathrm{id}$. Since $\mathsf{D}(f) \circ \eta = \eta \circ f$ and $\mathsf{D}^{k-1}(\mu) = \mathsf{D}(\mathsf{D}^{k-2}(\mu))$, $P_kP_k(w) = \eta\big(\eta(\mathsf{D}^{k-2}(\mu)\,\mathsf{D}^{k-1}(\mu)\,w)\big)$, which equals $P_k(w) = \eta(\mathsf{D}^{k-1}(\mu)w)$ only when $\mathsf{D}^{k-1}(\mu)w$ is already a point mass. (v) $\mathsf{D}^{k-1}(\mu) \circ \mathsf{D}^k(\eta) = \mathsf{D}^{k-1}(\mu \circ \mathsf{D}\eta) = \mathrm{id}$, so $P_k \circ \mathsf{D}^k(\eta) = \eta$. □

4 The shift law's defects

Theorem 4.1 (The unit defect)**.** *For every point (x, c),*

$$\lambda^\sigma \circ \eta_{\mathsf{C}_3X}(x, c) \;=\; (\mathrm{id} \times P)\big(\mathsf{C}_3(\eta_X)(x, c)\big).$$

At level k the right side is $\mathsf{D}^k(\eta_X)(c_k)$, the left side is $\delta_{c_k} = \eta(c_k)$, and pooling one depth at a time passes through the stages $s_j = \mathsf{D}^{k-j}(\eta_{\mathsf{D}^jX})(c_k)$, $j = 0, \ldots, k$, from s_0 (right side) to s_k (left side). In the shadow each hypothesis $(\theta, \mathsf{D}(\eta)\rho)$ is pooled to $(\theta, \delta_\rho) = K^{[1]}(\theta, \rho)$.

Proof. The components of the two sides are δ_{c_k} and $\mathsf{D}^k(\eta)(c_k)$ by the definitions. By lemma 3.2(v), $P_k(\mathsf{D}^k(\eta)c_k) = \delta_{c_k}$. One pooling at depth j is $\mathsf{D}^{k-j-1}(e)$, and $\mathsf{D}^{k-j-1}(e) \circ \mathsf{D}^{k-j}(\eta_{\mathsf{D}^jX}) = \mathsf{D}^{k-j-1}(e \circ \mathsf{D}(\eta_{\mathsf{D}^jX})) = \mathsf{D}^{k-j-1}(\eta_{\mathsf{D}^{j+1}X})$, which is $s_j \mapsto s_{j+1}$. In the shadow, $\mathsf{Sh}(\eta)$ sends (θ, ρ) to $(\theta, \mathsf{D}(\eta)\rho)$, and $e(\mathsf{D}(\eta)\rho) = \delta_\rho$. □

So the unit defect at level k is a chain of k elementary poolings. Stage s_j inserts the unit at depth j: the posterior is held as inherited spread below depth j and as a committed point above it. The draft's "crossover maps" compared c_k with its image; the stages are the correct intermediate objects, and its "coherence 2-cells" between levels are the whiskerings $\mathsf{D}^{k-j-1}(e)$.

Theorem 4.2 (The multiplication defect)**.** *For $M' = \sum_\ell \nu_\ell \delta_{M_\ell} \in \mathsf{D}^2\mathsf{C}_3X$,*

$$\lambda^\sigma \circ \mu_{\mathsf{C}_3X}(M') = \beta\big(\mathsf{D}(\lambda^\sigma)M'\big), \qquad \mathsf{C}_3(\mu_X) \circ \lambda^\sigma_{\mathsf{D}X} \circ \mathsf{D}(\lambda^\sigma_X)(M') = \beta\big(\mathsf{D}(\mathrm{id} \times P)\,\mathsf{D}(\lambda^\sigma)M'\big).$$

So the left side mixes the groups' images, and the right side mixes the groups' images after pooling each one.

Proof. The first equation is affinity of λ^σ. For the second, the level-k component of $\lambda^\sigma_{\mathsf{D}X}(\mathsf{D}(\lambda^\sigma)M')$ is $\sum_\ell \nu_\ell \delta_{z_\ell}$, where z_ℓ is the level-k component of $\lambda^\sigma(M_\ell)$, and $\mathsf{C}_3(\mu_X)$ applies $\mathsf{D}^k(\mu_X)$, which sends δ_z to $\delta_{\mathsf{D}^{k-1}(\mu)z} = P_k(z)$ by lemma 3.2(iv). The shadow is the same computation one level down, and the first components agree by (Co). □

Corollary 4.3 (Why the orientations clash)**.** *If P were affine, $\beta \circ \mathsf{D}(P) = P \circ \beta$, and the multiplication defect would read "right = P(left)", the same direction as the unit defect, "left = P(right)". P is not affine (lemma 3.2(iii)), and the two defects point opposite ways. The tower paper's exchanged pair [1, Prop. 13.17] is the extreme case: there the right side of (M) equals P applied to the right side of (U), which equals the left side of (U), and the left side of (M) equals the right side of (U). So no partial order orients both defects, and every order that reads pooling monotonically orients them oppositely [1, Thm. 13.15].*

Proposition 4.4 (When the defects vanish)**.** *(i) (U) holds at level k iff c_k is an iterated point mass of depth k (δ_x, δ_{δ_x}, $\delta_{\delta_{\delta_x}}$); it holds in the shadow iff every hypothesis in the support of Σ has a point-mass predictive.*

(ii) (M) holds at M' iff, within every group M_ℓ, all samples share their level-k context components and these are iterated point masses of depth $k-1$; the shadow never obstructs (M).

Proof. (i) $\delta_{c_k} = \mathsf{D}^k(\eta)c_k$ iff c_k is fixed by every inserted unit, i.e. an iterated point mass. (ii) By theorem 4.2 the two sides agree iff $P_k(z_\ell) = z_\ell$ for every group, and $z_\ell = \sum_i a_i \delta_{c_{i,k}}$ is fixed by $\eta \circ \mathsf{D}^{k-1}(\mu)$ iff it is a point mass whose atom is fixed by $\mathsf{D}^{k-1}(\mu)$ followed by η, which is the stated condition. In the shadow, e acts on $K^{[1]}$-images, which are already point masses. □

These replace the draft's identity criteria, which were stated for the quantiser.

5 The averaging law

The averaging law's only defect, (Cm), is pooling as well, of a different coordinate: at every level and in the shadow, its left side is its right side with each sample's context label e_i replaced by the pooled label $\bar{e} = \sum_i a_i e_i$, a barycenter, and this is a strict partial order natural in X [1, Thm. 13.16]. With one defect there is nothing to clash with, so all of $\bar{\lambda}$'s defects are 2-cells of one orientation. The draft's specialisation to the averaging law, a constant map onto the uniform law on the shadow's atoms, is not label pooling and is withdrawn.

6 What the draft got right and wrong

Object	Draft	This note
crossover maps χ_k	pairs (c_k, λ^σ_k), of different types ($\mathsf{D}^k X$ against $\mathsf{D}^{k+1} X$)	the stages $s_j = \mathsf{D}^{k-j}(\eta_{\mathsf{D}^j X})(c_k)$ between the two sides of (U)
coherence 2-cells α_k	$\chi_k \Rightarrow \chi_{k+1} \circ \delta$, between transformations with different codomains	the whiskerings $\mathsf{D}^{k-j-1}(e)$, one per depth
3-cells β_k	nearest-atom quantisers; $\beta_3 = \beta_S$	the pooling idempotent e, a 2-cell of **Cat**; no 3-cells are needed
coherence	"whiskering, pentagon, unit, vertical", asserted	idempotence of e, naturality, commutation with kernels, non-affinity: proved and checked
decomposition	not stated as an equation	theorems 4.1 and 4.2: exact factorisations
averaging law	constant map to a uniform law	label pooling [1, Thm. 13.16]

A tricategory is the wrong setting. **Cat** is a 2-category, natural transformations such as e are its 2-cells, and the defects are equations between composites of 1-cells and e. The axioms of a tricategory (associators, pentagonators and their modifications [4]) do not arise.

7 Relation to open problems

Weak distributive laws. $e = \eta_{\mathsf{D}} \circ \mu$ is a split idempotent, the kind that weak distributive laws split [5, 2]. Theorem 4.1 gives the shift law's unit axiom the shape of a weakened unit axiom, holding up to the idempotent; but its multiplication axiom fails too (theorem 4.2), so the shift law is not a weak law in the Street–Böhm sense, as the tower paper notes [1, Rem. 8.9]. Whether a mixed analogue of weak laws, weakening the monad axioms of a monad–comonad law by a split idempotent of the monad, makes the shift law a law is a precise question this note raises and does not answer.

Open problem 3. For the shift law it is closed negatively for partial orders [1, Prop. 13.17]; corollary 4.3 explains why: one idempotent, applied outside a mixture in (U) and inside it in (M). For the averaging law the open part is coherence in an order-enriched setting [1, Thm. 13.16].

Open problem 4. e is a non-identity 2-cell, but of **Cat**, which already has them. Open problem 4 asks for a 2-category whose 2-cells sit between the maps of the base, with a 2-monad extending D; e suggests what those 2-cells should compare, and nothing here constructs the 2-monad.

The Giry monad. Lemma 3.2 and theorems 4.1 and 4.2 use only the monad laws, the affinity of λ^σ and of Kleisli extensions, and the definitions. They therefore hold verbatim for any monad on a category where the shift law can be defined; for the Giry monad [3] the algebra carries over, and the measurability of the maps involved is not checked here.

8 Exact checks and audit

`check_pooling_cells.py` uses the definitions of `check_tower.py` verbatim and checks, in exact rational arithmetic:

P1 e is idempotent, preserves the barycenter, is natural for an injective and a non-injective map, and commutes with $\mathsf{D}(k^{\#})$ for 200 random kernels on a three-point base (0 failures);
P2 e is not affine;
P3 theorem 4.1 in every component on 200 random points, with all stages s_j;
P4 theorem 4.2 on 200 random two-level samples with up to three groups;
P5 proposition 4.4, in both directions, on random and planted cases;
P6 the exchanged pair of [1, Prop. 13.17];
P7 the closed form $P_k = \eta \circ \mathsf{D}^{k-1}(\mu)$, and non-idempotence of P_k for $k \geq 2$ (198 of 200 cases).
It also audits the draft. The nearest-atom map moves the barycenter of $\delta_{(9/10,1/10)}$ to $(\frac{3}{4}, \frac{1}{4})$, gives different values on $\delta_{(1/2,1/2)}$ when its two atoms are listed in opposite orders, and fails to commute with $\mathsf{D}(k^{\#})$ in 195 of 400 random cases on a three-point base. The draft's second and third scripts reproduce their stated hashes; the first gives a different hash from the one stated, and its β_1 returns the empty distribution on $\mathsf{D}(\eta)\rho$. In the second and third scripts the four "coherence" results are written as the constant `True`, and the averaging-law test checks a constant function against its own value. All checks of this note pass; the hashes of the script and of the draft's scripts are in `FROZEN.txt`.

References

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